

Manual's user for the Digital Transmission AWGN bandlimited channel simulator – version 3.1

Simulator Objectives

Analyzing the performances of PAM2 or PAM4 modulated signal in a bandlimited AWGN channel with the option of using a zero forcing equalizer or MMSE equalizer.

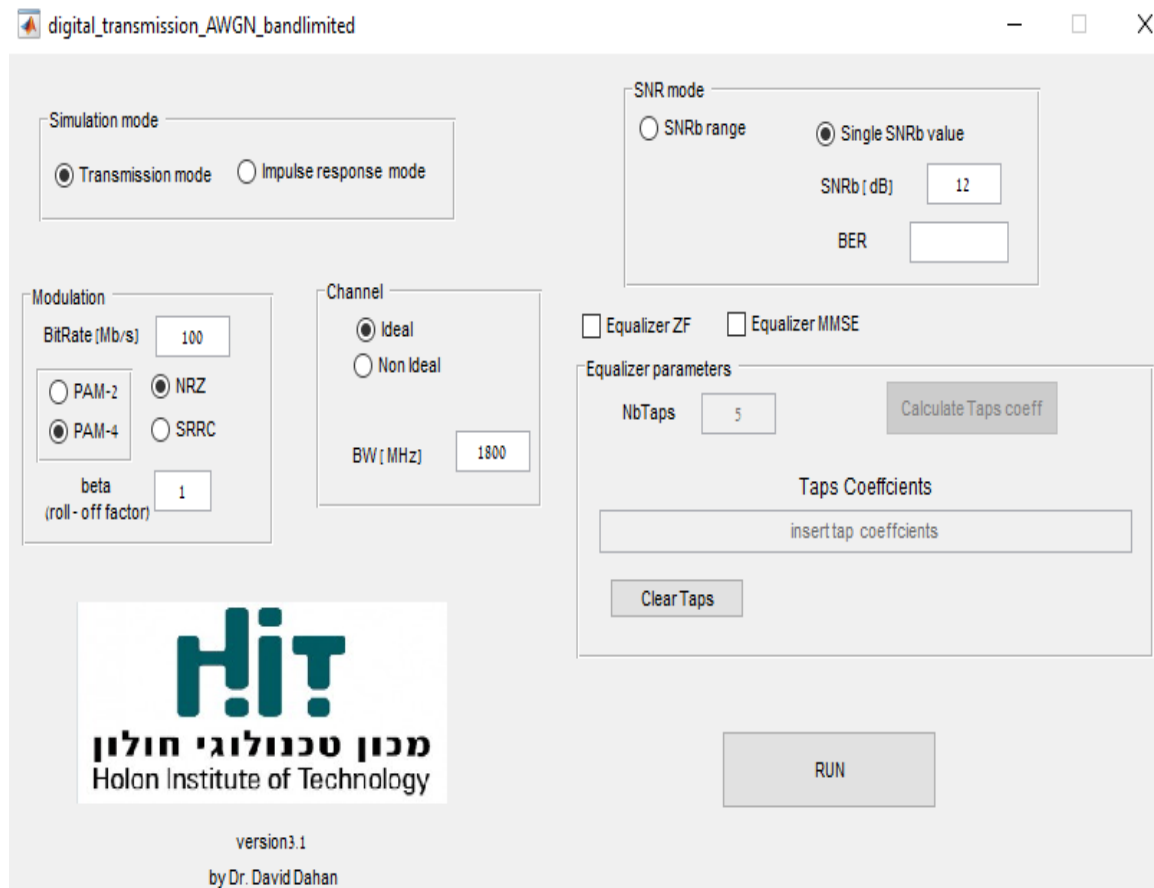
Running instructions

The simulator runs in MATLAB environment and requires to use the following toolboxes:

- Communication toolbox
- Signal processing toolbox

In order to open the simulator GUI, run the file `digital_transmission_AWGN_bandlimited.m` (not the file `digital_transmission_AWGN_bandlimited.fig` !!!)

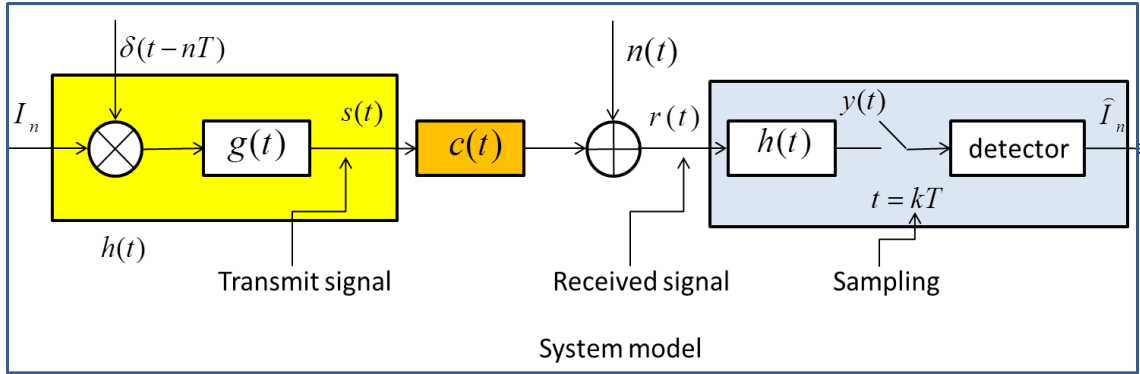
Simulator GUI



Simulator parameter configurations

1) Simulation mode

The following system is simulated :



Where $s(t)$ is a M-PAM signal at a symbol rate of $S_R = \frac{1}{T_s}$:

$$s(t) = \sum_n I_n g(t - nT_s)$$

Where $g(t)$ is the basic pulse shape and $I_n \in \{-1, 1\}$ for PAM2 or $I_n \in \{-3, -1, 1, 3\}$ for PAM4 with equal transmission probability. Gray coding is used for PAM4 modulation format.

The digital signal is transmitted into an AWGN channel with a channel transfer function and the received signal is $r(t) = s(t) * c(t) + n(t)$, where $c(t)$ represents the impulse channel response and $n(t)$ is a zero mean stationary white Gaussian noise with a spectral density function $\phi_{nn}(f) = \frac{N_0}{2}$.

The overall system impulse response is defined as $x(t) = g(t) * c(t) * h(t)$

The receiver is based on the minimum distance criterion :

$$\|y(kT_s) - \hat{I}_k\| = \min_{j=1:M} (\|y(kT_s) - I_j\|)$$

Where $y(t) = r(t) * h(t)$ with $h(t)$ is the receiver matched filter to $g(t)$ and $\hat{I}_k$ denotes the decided symbol at time kT_s .

When the equalizer button is selected, a Zero Forcing equalizer filter or an MMSE equalizer filter is placed after the matched filter $h(t)$ and the impulse response of the equalizer is $e(t)$.

2) Modulation

The signal bit rate can be defined in this GUI section. It is possible to choose either PAM2 or PAM4 modulation format.

The transmitter filter response $g(t)$ can be chosen to be either a NRZ rectangular pulse or a square root raised cosine pulse. In this later case the roll off factor of the pulse (beta) is required. The receiver filter impulse response, $h(t)$ is set as being the matched filter of $g(t)$

3) Channel

The channel bandwidth is defined in this section as BW [MHz]

The channel impulse response can be chosen as being:

a) Ideal : low pass rectangular response

$$C(f) = \begin{cases} 1 & \text{when } |f| \leq BW \\ 0 & \text{otherwise} \end{cases}$$

b) Non ideal

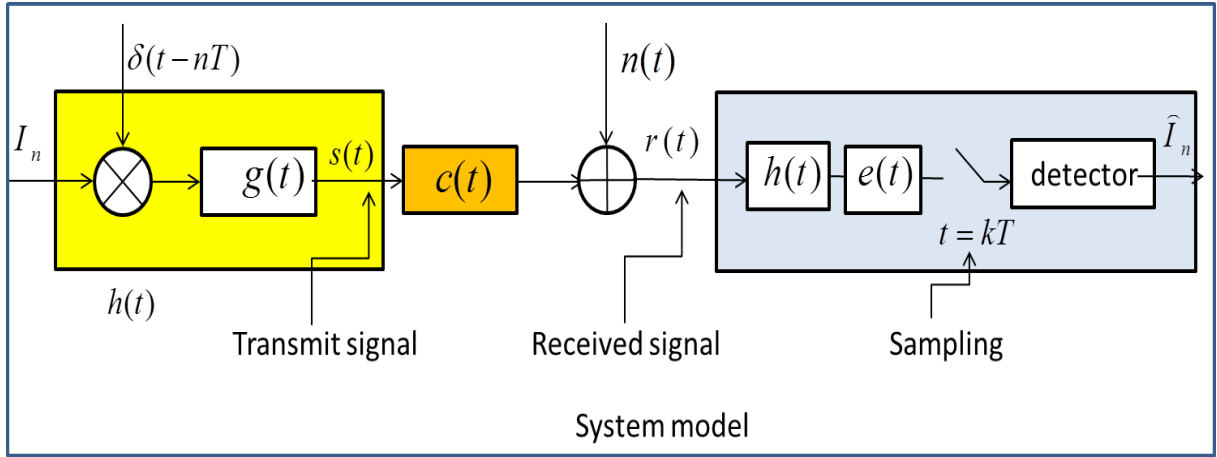
$$C_1(f) = C(f) \times \left(\frac{1 - 0.15 \exp(+j2\pi f \times 0.95T_0) + 0.79 \exp(-j2\pi f \times 4.1T_0)}{1 - 0.2 \exp(-j2\pi f \times 0.7T_0)} \right)$$

$$\text{where } C(f) = \begin{cases} 1 & \text{when } |f| \leq BW \\ 0 & \text{otherwise} \end{cases} \quad \text{and } T_0 = 350 \text{ ps}$$

4) Equalizer

Two types of equalizer can be selected : ZF equalizer or MMSE equalizer

When the equalizer mode is selected, a ZF or MMSE “analog” equalizer is set after the receiver matched filter, as shown below:



$e(t)$ is the impulse equalizer response. In the equalizer mode, an odd number of taps has to be

selected. Assuming $2N+1$ taps, we define $e(t) = \sum_{n=-N}^N d_n \delta(t-nT)$

4.1 : Calculate taps coefficients for ZF equalizer

When ZF equalizer is selected and when pressing calculate Taps coeff, the tap coefficient are calculated according to the zero forcing problem and displayed in the Taps coefficients field:

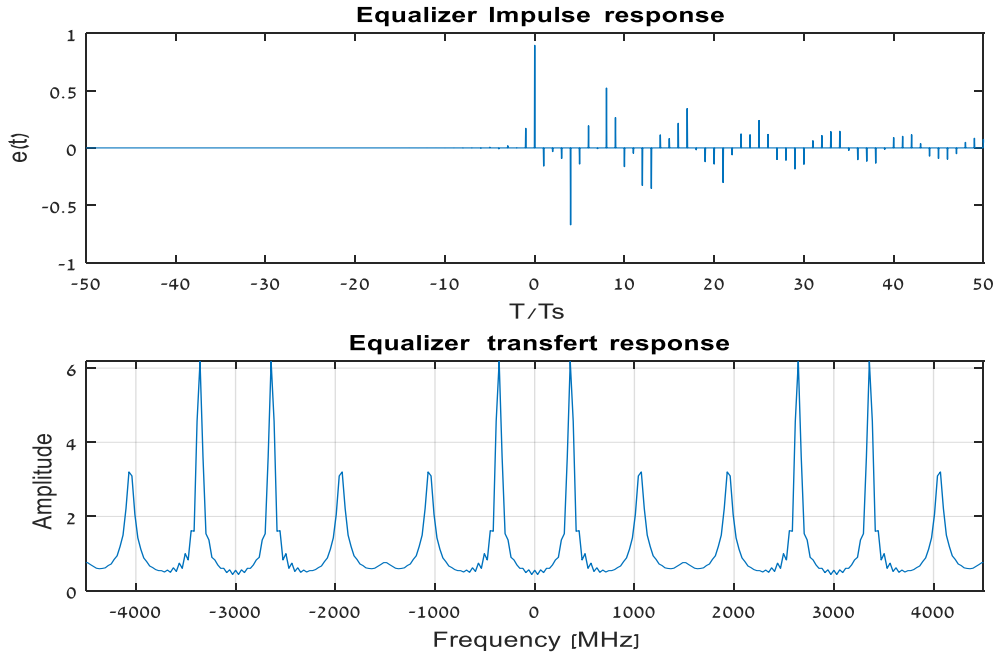
$$\vec{d} = \begin{bmatrix} d_{-N} \\ \vdots \\ d_{-1} \\ d_0 \\ d_1 \\ \vdots \\ d_N \end{bmatrix} = X^{-1} \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

Where

$$X = \begin{bmatrix} x(0) & x(-T) & \cdots & \cdots & x(-2NT) \\ x(T) & \ddots & \ddots & & \vdots \\ \vdots & x(T) & x(0) & x(-T) & \\ & & x(T) & x(0) & x(-T) \\ & & & x(T) & x(0) & x(-T) \\ \vdots & & & & \ddots & \ddots & \ddots \\ x(2NT) & \cdots & \cdots & \cdots & x(T) & x(0) \end{bmatrix}$$

With $x(t) = g(t) * c(t) * h(t)$

The impulse response of $e(t)$ as well as its transfer function is also plotted



4.2 : Calculate taps coefficients for MMSE equalizer

When MMSE equalizer is selected and when pressing calculate Taps coeff, the tap coefficient are calculated according to the MMSE problem and displayed in the Taps coefficients field :

$$\vec{e}_{MMSE} = R_{yy}^{-1} R_{ly}$$

Where

$$R_{yy} = \Gamma_{xx} + \frac{N_0}{2} I = \begin{bmatrix} \Gamma_{xx}(0) & \Gamma_{xx}(1) & \Gamma_{xx}(2) & \dots & \Gamma_{xx}(2N) \\ \Gamma_{xx}(1) & \Gamma_{xx}(0) & \Gamma_{xx}(1) & \dots & \Gamma_{xx}(2N-1) \\ \Gamma_{xx}(2) & \Gamma_{xx}(1) & \ddots & \ddots & \vdots \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ \Gamma_{xx}(2N) & \Gamma_{xx}(2N-1) & \dots & \Gamma_{xx}(1) & \Gamma_{xx}(0) \end{bmatrix} + \frac{N_0}{2} \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & 0 & \dots & 1 \end{bmatrix}$$

(2N+1)x (2N+1) covariance matrix of channel response
(We use that $\Gamma_{xx}(-k) = \Gamma_{xx}(k)$)

(2N+1)x (2N+1) Identity matrix

And

$$R_{fy} = \begin{bmatrix} x[N] \\ x[N-1] \\ \vdots \\ x[0] \\ \vdots \\ x[-N+1] \\ x[-N] \end{bmatrix}$$

The calculation of the taps coefficients assume the SNRb level displayed in the GUI field.

5) Simulation Mode

Two simulation modes can be selected :

a) Transmission mode

In this mode, when pressing RUN button, bits are generated randomly, modulated using either PAM2 or PAM4 modulation format at the bit rate defined by the field BitRate in the GUI and sent over the bandlimited AWGN channel with bandwidth defined by BW .

The BER calculation module compares the received bit stream to the sent bit stream. The BER is calculated by divided the number of error bits by the total number of sent bits. When the run button is pressed, 4096 randomly generated bits are sent through the channel and it is repeated 50 times in order to be able to detect BER level as low as 10^{-5} .

In this mode, 4 figures are displayed:

Figure 1 shows the 100 Mbit/s PAM2 signal and 100 Mbits/s PAM4 signal at the transmitter and the corresponding eye diagram at 100Mbit/s with NRZ line coding.

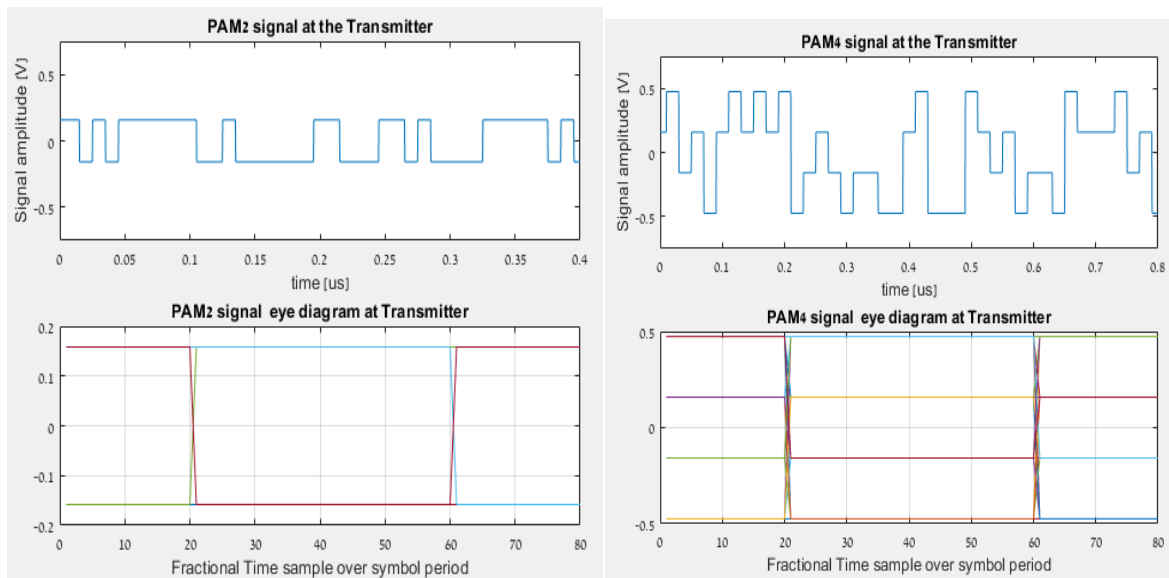


Figure 2 shows the power spectra density of the 100 Mbits/s PAM2 signal at the transmitter output

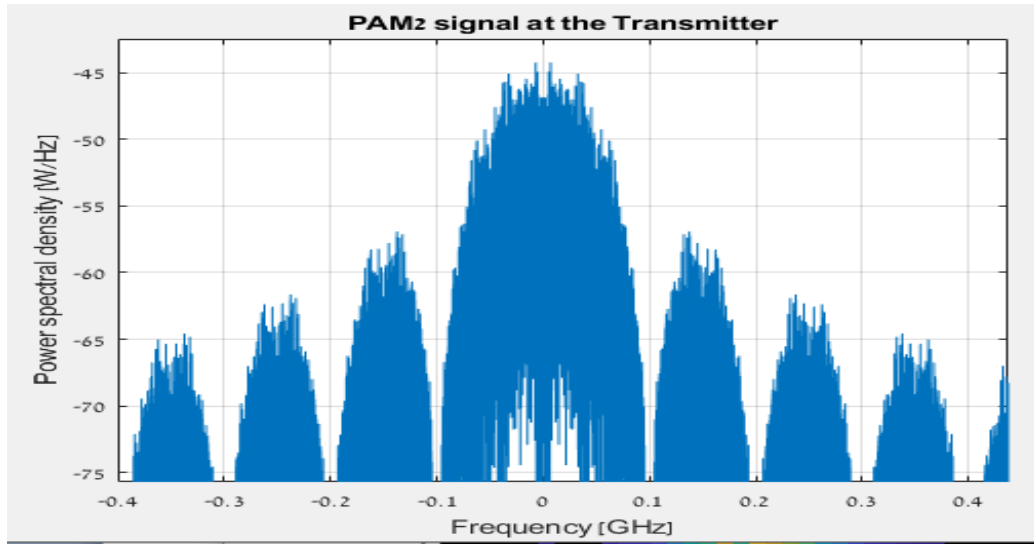


Figure 3 shows the received noisy 100 Mbits/s PAM2 and 100 Mbits/s PAM4 at the output of the receiver filter and its corresponding eye diagram

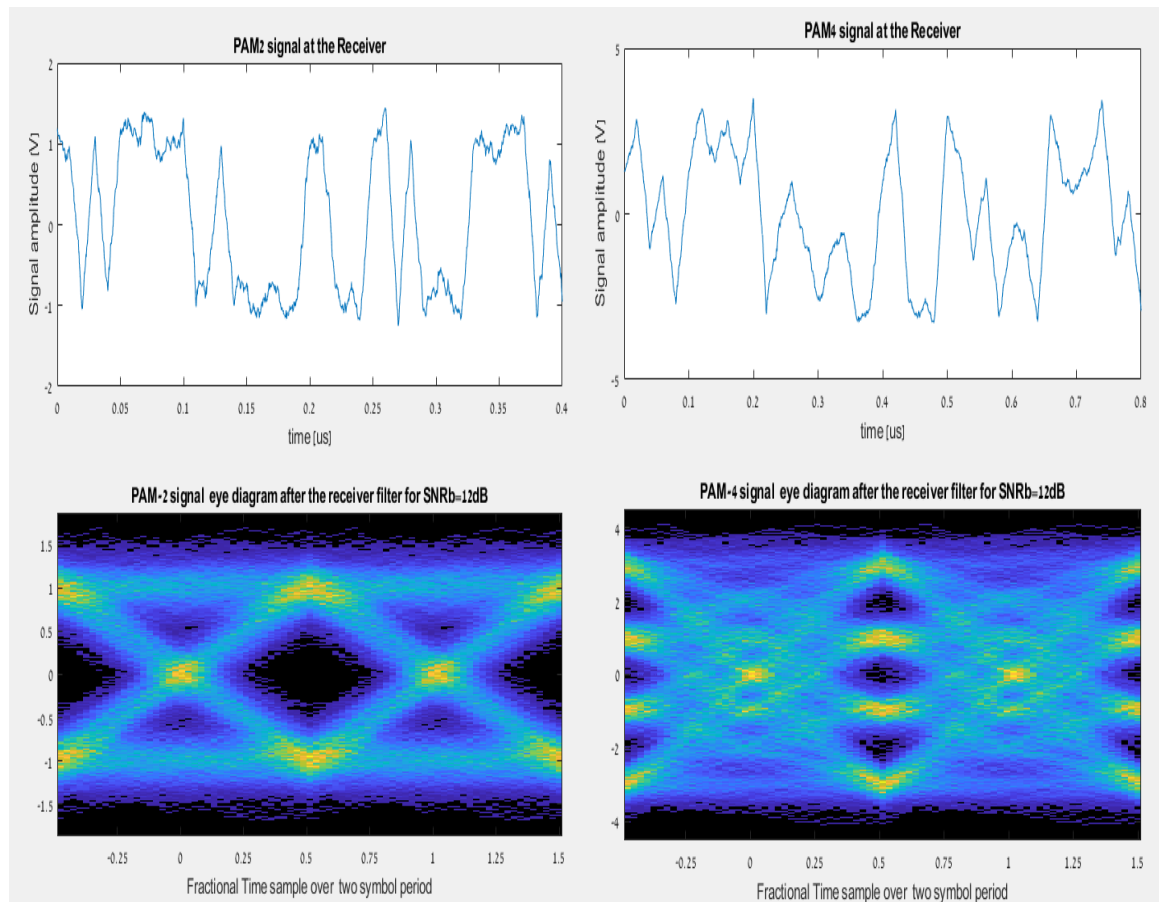
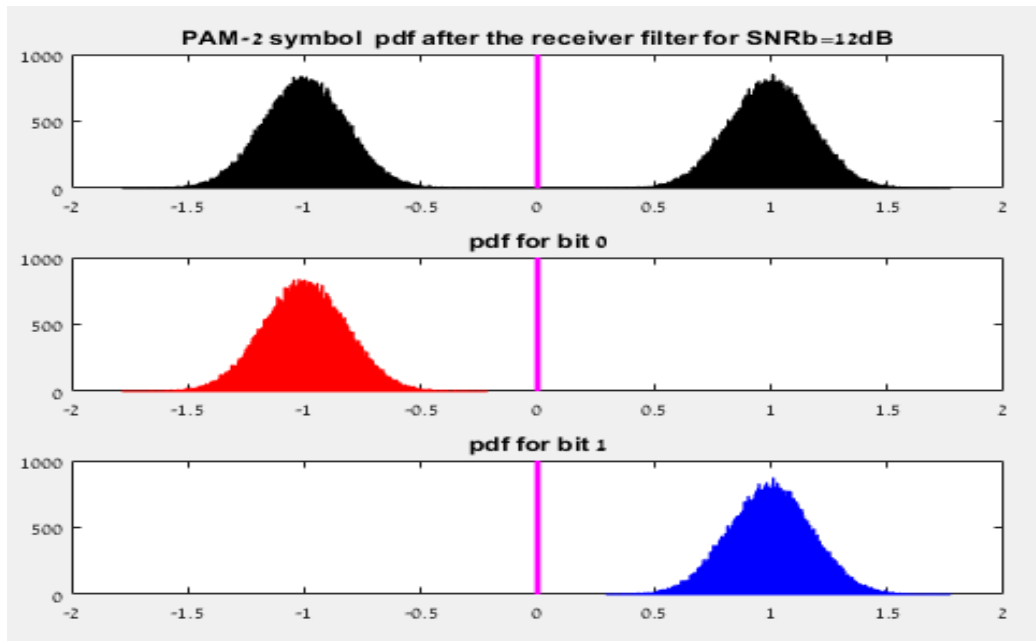


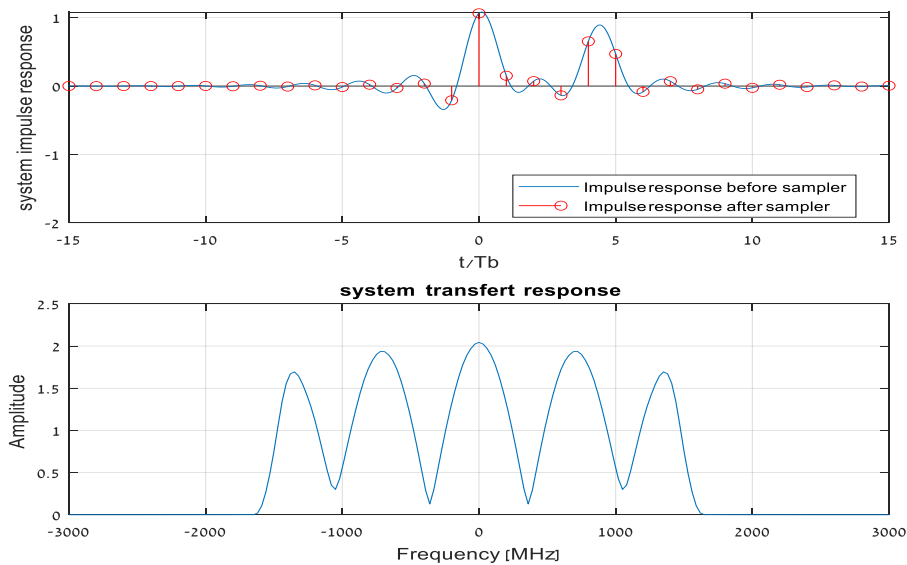
Figure 4 shows the total pdf of the received symbols (only for the case of PAM2 modulation) , the conditioned pdf for bit 1 and for bit 0



b) Impulse response mode

In this mode, when pressing RUN button, a single bit is sent in order to determine the system transfer function.

In this mode, the impulse response of the system, $x(t) = g(t) * c(t) * h(t)$, is displayed as well as its sampled impulse response after the sampler and the system transfer function $X(f)$



When the equalizer button ZF or MSSE is selected, $x(t) = g(t) * c(t) * h(t) * e(t)$ where the impulse response of the equalizer is $e(t)$ according to the ZF or MMSE coefficient calculation.

6) SNR Mode

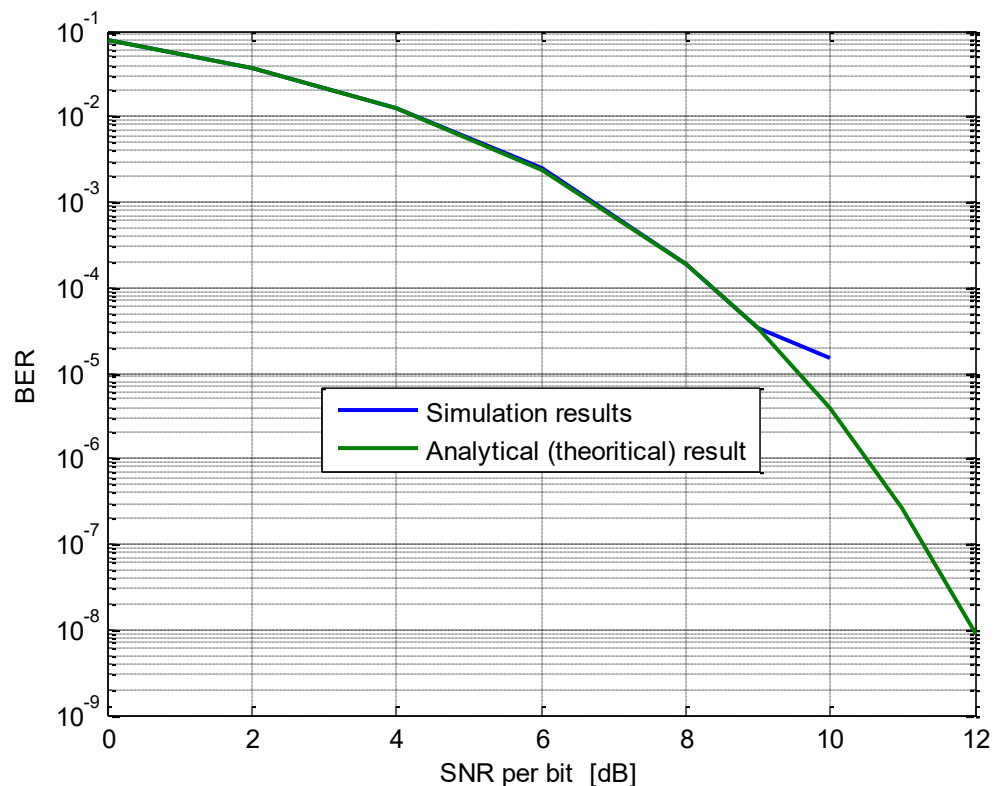
This mode is relevant when only when transmission mode is selected

6.1 : SNRb range :

In this mode the SNR_b range varies from 0 to 20 dB when selecting ideal channel or non ideal channel.

At the loop end, the BER vs SNR_b graph is displayed for the numerical BER (based on the simulation) and the analytical BER expression for a polar signal.

The numerical BER calculation is done by comparing the received bit stream to the sent bit stream. The BER is calculated by divided the number of error bits by the total number of sent bits.



6.2 : single SNRb value:

In this mode, the SNR_b is set to a single value and the result in BER is displayed in the GUI.